Shuxu "Sev" Huffman

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Education

Ph.D. **Université Paris-Saclay, France; Gallaudet University, USA** — Spring 2026 - Present
International Cotutelle | Computer Science; Human-Computer Interaction

M.Sc. **University of Washington, USA**, Computer Science and Engineering — Fall 2019 - Summer 2020

B.Sc. **University of Washington, USA**, Computer Science — Winter 2017 - Spring 2019

B.Sc. **Gallaudet University, USA**, Information Technology — Fall 2013 - Spring 2016
Magna Cum Laude; University Honors; Swim Team

Research Experience

Deaf-Centered AI | Sign Language AI

- Investigate how Deaf users experience and evaluate AI systems, including large language models and sign-language AI, with attention to translation quality, accessibility, multimodal communication, and failures involving spatial and embodied language.
- Conduct qualitative and longitudinal studies of real-world AI use, analyzing signer variation, cultural context, communication practices, and mismatches between model behavior and Deaf users' needs.
- Contributed to related work published at *CHI* and in *Computers & Education: Artificial Intelligence*.

DeafSpaceXR | Immersive Learning Systems

- Designed and implemented an Apple Vision Pro classroom prototype for Deaf learners using Swift and RealityKit, supporting configurable signer and instructional-content layouts, visual attention controls, embedded questions, and interaction logging.
- Led mixed-methods studies with Deaf educators and community members to investigate visual attention, spatial configuration, and accessibility in immersive learning environments.
- Published resulting work at *CHI* and *ASSETS*, including an Honorable Mention paper, and developed a broader Deaf-centered framework for immersive-system design.

DeafSwim | Accessible Sports Technology

- Designed and deployed a haptic communication system for Deaf open-water swimming, enabling a support kayaker to send directional cues to the swimmer through a smartwatch during long-distance swims.
- Evaluated the system in real-world open-water conditions, examining reliability, attention, and interdependence in safety-critical swimmer–kayaker communication. Published resulting work at *ASSETS*.

Teaching Experience

Adjunct Instructor, Gallaudet University | ITS 395/595 - AI Software Development Tools — Jan. 2024 - May 2025

- Designed and taught an introductory AI software development course covering how large language models are integrated into applications, including embeddings, similarity-based retrieval, vector databases, and retrieval-augmented generation.
- Guided students through the end-to-end development of AI applications using Python, Flask, and OpenAI APIs, connecting core AI concepts with practical software engineering and responsible system design.

Teaching Assistant, University of Washington — 2018 - 2020

- CSE 120 - Computer Science Principles | Winter 2018
- CSE 351 - The Hardware/Software Interface | Fall 2018, Winter 2019
- CSE 340 - Interaction Programming | Winter 2020, Spring 2020

Selected Publications

Conference Papers

C1. **Huffman, Shuxu**, Glasser, A., Hayes, C., Vogler, C. & Kushalnagar, R. *"We need a vision first": Speculating Deaf-Centered Immersive Classrooms* in *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems* (Association for Computing Machinery, Barcelona, Spain, 2026). ISBN: 979-8-4007-2281-3/2026/04. https://dl.acm.org/doi/10.1145/3772363.3798360.

C2. **Huffman, Shuxu** & Kushalnagar, R. *Situated Minimalism: When One Button Becomes Relational Access in Deaf Open Water Swimming* in *Proceedings of the 28th International ACM SIGACCESS Conference on Computers and Accessibility* (Association for Computing Machinery, New York, NY, USA, 2026), 1–4. https://doi.org/10.1145/3797867.3841196.

C3. **Huffman, Shuxu**, Okosi, M., Merino, J., DeVries, P., Bouchara, T., Vogler, C., Glasser, A. & Kushalnagar, R. *XR for Whom? A Critical Review of Extended Reality Research on Deaf and Hard of Hearing Users* in *Proceedings of DeafTech 2026: The International Deaf Technology Conference* (Vienna, Austria, 2026). https://hal.science/hal-05628514v1/document.

C4. **Huffman, Shuxu**, Okosi, M., Merino, J., Bouchara, T., Glasser, A., Vogler, C. & Kushalnagar, R. *Bigger than the EAR BOX: A Theory-Grounded Review of XR Accessibility Research for Deaf and Hard of Hearing Communities* in *Proceedings of the 28th International ACM SIGACCESS Conference on Computers and Accessibility* (Association for Computing Machinery, New York, NY, USA, 2026), 1–17. https://doi.org/10.1145/3797867.3828978.

C5. **Huffman, Shuxu**, South, L., Buckman, M., Kushalnagar, R., Ortega, F. R. & Glasser, A. *Reclaiming VR Design Authority: Deaf Signers Shaping Immersive Classrooms* in *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems.* **Honourable Mention Award (Top 5% of submissions)** (Association for Computing Machinery, Barcelona, Spain, 2026). ISBN: 979-8-4007-2278-3/2026/04. https://doi.org/10.1145/3772318.3790499.

C6. Perdriau, C., **Huffman, Shuxu**, Watkins, A. & Lewis, C. M. *"I can't do four things at once": Deaf Students' Experiences With and Recommendations for Improving the Accessibility of CS Lectures and Live Coding* in *Proceedings of the 2026 ACM Conference on International Computing Education Research Vol. 1* (2026), 274–288. https://dl.acm.org/doi/10.1145/3765964.3811652.

C7. **Huffman, Shuxu**, Angelini, R., Kushalnagar, R. & Spiel, K. *"It only needs to work for one of us": Rethinking DIY Deaf Tech Through Situated Co-Design* in *The 27th International ACM SIGACCESS Conference on Computers and Accessibility* (Association for Computing Machinery, Denver, CO, USA, 2025). ISBN: 979-8-4007-0676-9/2025/10. https://doi.org/10.1145/3663547.3749828.

C8. **Huffman, Shuxu**, Chen, S., Mack, K. A., Su, H., Wang, Q. & Kushalnagar, R. *"We do use it, but not how hearing people think": How the Deaf and Hard of Hearing Community Uses Large Language Model Tools* in *Extended Abstracts of the 2025 CHI Conference on Human Factors in Computing Systems* (Association for Computing Machinery, Yokohama, Japan, 2025). ISBN: 979-8-4007-1395-8. https://doi.org/10.1145/3706599.3719785.

Journal Papers

J1. Chen, S., **Huffman, Shuxu**, Zhu, Q., Su, H., Wang, Q. & Kushalnagar, R. Exploring the design of LLM-powered question generation for deaf and hard of hearing learners. *Computers and Education: Artificial Intelligence,* 100615. https://www.sciencedirect.com/science/article/pii/S2666920X26000779 (2026).

Workshop Papers

W1. Angelini, R., De Meulder, M., **Huffman, Shuxu**, Kushalnagar, R., Glasser, A. & Suchanek, O. *Sign-Up on Deaf Technologies: Reframing Access, Interaction, and Design* in *Proceedings of the Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems* (2026), 1–3. https://dl.acm.org/doi/10.1145/3772363.3778788.

W2. Mack, K. A. & **Huffman, Shuxu**. *Why Researchers Working with the Deaf Community Should Learn ASL* in *CHI '20: International Conference on Human Factors in Computing Systems* (Honolulu, HI, USA, 2020). https://kmack3.github.io/Mack_and_Tian.pdf.

Employment

Microsoft Corporation, Software Engineer 2 — Sept. 2020 - July 2026
Azure OneBranch; Office AI

- Built and owned production software across Azure OneBranch and Office AI, including Live Components used across Microsoft Loop, Teams, Outlook, and Office for the Web.
- Designed testing and deployment infrastructure in TypeScript, JavaScript, and C#, improving reliability, performance, security, and maintainability.
- Mentored three undergraduate software engineering interns, providing technical guidance, project feedback, and career support; two later returned to Microsoft as full-time employees.
- Led accessibility-focused prototyping and hackathon projects for Deaf and Hard of Hearing users, including a 1st-place Diversity Hack project and collaboration with DeafSpotlight.

Internships

- Software Engineer Intern, Bloomberg LP — Summer 2020
- Software Engineer Intern, Tableau Software — Summer 2019
- Software Engineer Intern, Microsoft Corporation — Summer 2016, 2017, 2018, and Fall 2019
- Explorer Intern, Microsoft Corporation — Summer 2015

Awards & Honors

- ACM CHI | 2 Special Recognitions for Outstanding Reviews — 2026
- National Science Foundation CSGrad4US Graduate Fellowship — 2024
- Microsoft Office | 1st place in Diversity Hack — 2022
- Tableau Data Viz competition | 1st place in Intern Summit — 2019
- Microsoft | 1st place in HoloHack Challenge — 2016
- North Eastern Athletic Conference | Scholar-Athlete Award — 2014-2016
- Northrop Grumman | Scholarship Award — 2014
- Nippon Foundation | Sasakawa International Scholarship — 2014-2016

Programming Languages

C# Java Python TypeScript Swift JavaScript | HTML | CSS

Natural Languages

English (Fluent) Chinese (Native) French (DELF B1)
American Sign Language (Fluent) Chinese Sign Language (Proficient)